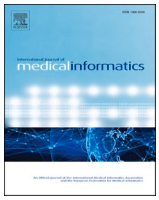
International Journal of Medical Informatics 188 (2024) 105466



Contents lists available at ScienceDirect

International Journal of Medical Informatics

journal homepage: www.elsevier.com/locate/ijmedinf





Analysing disease trajectories in a cohort of 71,849 Patients: A visual analytics and statistical approach

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ARTICLE INFO

Keywords:
Disease trajectories
Data mining
Data exploitation
Graphical analysis
Interactive visualization

ABSTRACT

Background: Disease trajectories have become increasingly relevant within the context of an aging population and the rising prevalence of chronic illnesses. Understanding the temporal progression of diseases is crucial for enhancing patient care, preventive measures, and effective management.
Objective: The objective of this study is to propose and validate a novel methodology for trajectory impact analysis and interactive visualization of disease trajectories over a cohort of 71,849 patients.
Methods: This article introduces an innovative comprehensive approach for analysis and interactive visualization of disease trajectories. First, Risk Increase (RI) index is defined that assesses the impact of the initial disease diagnosis on the development of subsequent illnesses. Secondly, visual graphics methods are used to represent cohort trajectories, ensuring a clear and semantically rich presentation that facilitates easy data interpretation.
Results: The proposed approach is demonstrated over the disease trajectories of a cohort comprising 71,849 patients from Tolosaldea, Spain. The study finds several clinically relevant trajectories in this cohort, such as that after suffering a cerebral ischemic stroke, the probability of suffering dementia increases 10.77 times. The clinical relevance of the study outcomes have been assessed by an in-depth analysis conducted by expert clinicians. The identified disease trajectories are in agreement with the latest advancements in the field.
Conclusion: The proposed approach for trajectory impact analysis and interactive visualization offers valuable graphs for the comprehensive study of disease trajectories for improved clinical decision-making. The simplicity and interpretability of our methods make them valuable approach for healthcare professionals.

1. Introduction

In this study, the term "disease trajectories" refers to the sequential onset of diseases over time on the same patient. They have become a crucial area of research, especially with the ageing population and the rising prevalence of chronic diseases [1]. Disease trajectory analysis consists in grouping patients with similar disease patterns and examining how their health status changes over time. By examining disease trajectories, researchers can identify risk factors, gain insights into the factors that contribute to disease progression and determine the most effective interventions. In the same way, it can provide valuable information for the policy makers to prevent them or to start early care at the population level.

Despite the promise of trajectory analysis, the approach used to analyse disease trajectories need to be rigorous and reproducible to ensure the reliability of the results, and to allow confirmation of the findings by third parties. Furthermore, results of current approaches can be complex and challenging to interpret, particularly when multiple diseases are involved.

In this paper, we present an analysis of diseases trajectories over a cohort of 71,849 patients from Tolosaldea (Spain), following an innovative approach that uses a combination of trajectory impact analysis

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https://doi.org/10.1016/j.ijmedinf.2024.105466
Received 16 February 2024; Received in revised form 8 April 2024; Accepted 22 April 2024
Available online 1 May 2024
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and an interactive visualization software tool. Its goal is to provide a prioritised, ordered, and easy-to-understand description of disease trajectories, allowing clinicians and researchers to identify key points in disease progression and risk factors. We apply our analysis to a specific disease or set of diseases and examine how different disease diagnoses follow one another over time.

Our proposed computational platform for analysing disease trajectories includes both statistical and visual methods. It makes use of the trajectory impact analysis to identify the most significant trajectories, and an interactive visualization software library to display them in a prioritized, ordered, and simple-to-understand manner.

We demonstrate how these methods can be used to identify key precursor diseases. Moreover, it could also be used to assess increases in the probabilities of adverse events within the same disease.

2. State of the art

Traditionally, combining data mining with graphical methods has been beneficial and used in many sectors besides medicine [2], such as security, finance, and business [3,4].This is done with the aim of synthesizing information and deriving insights from massive, dynamic, ambiguous, and often conflicting data. This allows to detect the expected and discover the unexpected; providing timely, defensible, and understandable assessments; and communicating these assessments effectively for action [5,6]. Therefore, several medical studies incorporate data mining into visualizations, such as combining machine learning techniques with visual representations [7,8], map plots for visualization of population statistics [9,10], or combining statistical approaches with trajectories graph visualizations [11,12,13,14,15,16,17], the focus of this study.

In a study presented in 2019 [11] they analysed the disease trajectories of 10.4 million patients who were admitted between 1980 and 2010 to one of 350 hospitals. In total, 300 clinically relevant trajectories were identified, among which one trajectory stood out, which yielded an unprecedented outcome hitherto unreported in scientific literature. The study was supported by graphical representations of the trajectories by directed graphs, which facilitated intuitive insights, such as the average age at diagnosis using the node colouring or the death with a black square. Two years after, they presented a follow-up study [12] in which they focused on the temporal correlation between diseases and mortality. They identified 300 mortality trajectories, based on 118 diseases.

Two trajectory analysis studies have been presented on the population of Denmark [13,14], using in the first 14.9 years of record and in the second 25. In the first study [13], they managed to identify 1,171 significant trajectories, while in the second [14] they proposed a framework (called Danish Disease Trajectory Browser, DTB), where users can identify pairs of diagnoses with a statistically significant direction and combine them to obtain linear disease trajectories.

Other studies have focused on a specific disease. On one hand, depression has been analyzed [15] in the UK Biobank cohort, identifying significant associations between depression and 129 medical conditions. Additionally, three depression-mortality trajectories were identified, offering insights into mortality risks. In the other hand, type 2 diabetes mellitus has been analysed in a study carried out in South Korea [16]. The study identified 76,423 significant patterns, with the top 30 showing microvascular, macrovascular, and miscellaneous complications.

In our study, we performed comparable computational experiments but with variations in three aspects: data used, statistical techniques applied, and visualization methods employed. This dataset has not been utilized for this specific purpose in any previous studies, enabling the discovery of local patterns or obtaining a comprehensive local overview of disease progressions. Additionally, we offer interactive visualizations that empower users to examine patterns and validate clinical suspicions.

3. Materials and methods

3.1. Material

For this study, all patient interactions with Asuncion Klinika (Tolosa, Spain) from 2000-01-01 to 2021-08-11 were collected, totalling 526,597 interactions involving 115,545 different patients. These interactions included any medical episode or incident experienced by patients. Patients with only one interaction with the hospital were excluded from the analysis because a trajectory requires at least two records to capture the progression from one medical condition to another. As a result, our analysis focuses on 71,849 patients (36,489 female) who, on average, had 8.9 interactions in their medical history reports. Furthermore, it should be noted that deaths are not recorded in hospital data unless they occur during an admission or stay, and hence, dead has not been used in this study.

3.2. Data preparation

In preparation for the analysis, a data preparation process was executed, comprising three fundamental steps: Diagnostic code harmonization via BDCAP, patient vector representations, and trajectory data analysis.

3.2.1. Diagnostic code harmonization – BDCAP

In each interaction with the patient, the clinician recorded an ICD code associated with the diagnosed disease or health-related problem. Depending on the year in which the interaction took place the coding of the diagnosis was either ICD-9[1] or ICD-10[2]. These code sets have a totally different coding, and a clear one-to-one correspondence between them is missing in the publications of the regulatory agencies, making it challenging to link the same diagnoses that have occurred at different times. Moreover, the ICD coding is excessively specific, differentiating between very similar diagnoses.

To address this problem, we used the BDCAP (for its acronym in Spanish, “Base de Datos Clínicos de Atención Primaria” which means “Primary Care Clinical Database”) coding system [18], which harmonizes all the codes in the database, effectively matching ICD-9 and ICD-10. Moreover, BDCAP coding system groups diseases into fewer categories, resulting in 645 different diagnoses in our cohort.

3.2.2. Patient vector representations

For each patient, we created a vector representation containing all interactions had throughout his life, together with the age at which each interaction occurred. For example, if a patient had two interactions, at 47.2 years of age and at 51.7 years of age, respectively, the vector would look like this, with each interaction represented by its BDCAP code extracted for each contact:

[[47.2,’A98′], [51.7, ’D06′]]

By organizing the patient interactions in chronological order, we can identify the sequence in which they occurred and analyse potential causal relationships between different interactions, enabling us to conduct a population study.

3.2.3. Trajectory data

To analyse the trajectories between each disease, we paired all different diagnose codes used in the cohort (645), resulting in a total of 207,690 pairs of diagnose codes, regardless of order. Given the large number of code pairs and the low incidence of many of them, a filter was applied requiring at least 5 patients with both diagnoses for the code

[1] https://www.sanidad.gob.es/ca/estadEstudios/estadisticas/estadisticas/estMinisterio/SIAP/Mapeo_CIE9MC_BDCAP_31_07_2020.xlsx.

[2] https://www.sanidad.gob.es/ca/estadEstudios/estadisticas/estadisticas/estMinisterio/SIAP/Mapeo_CIE10_BDCAP_11_02_2019.xls.

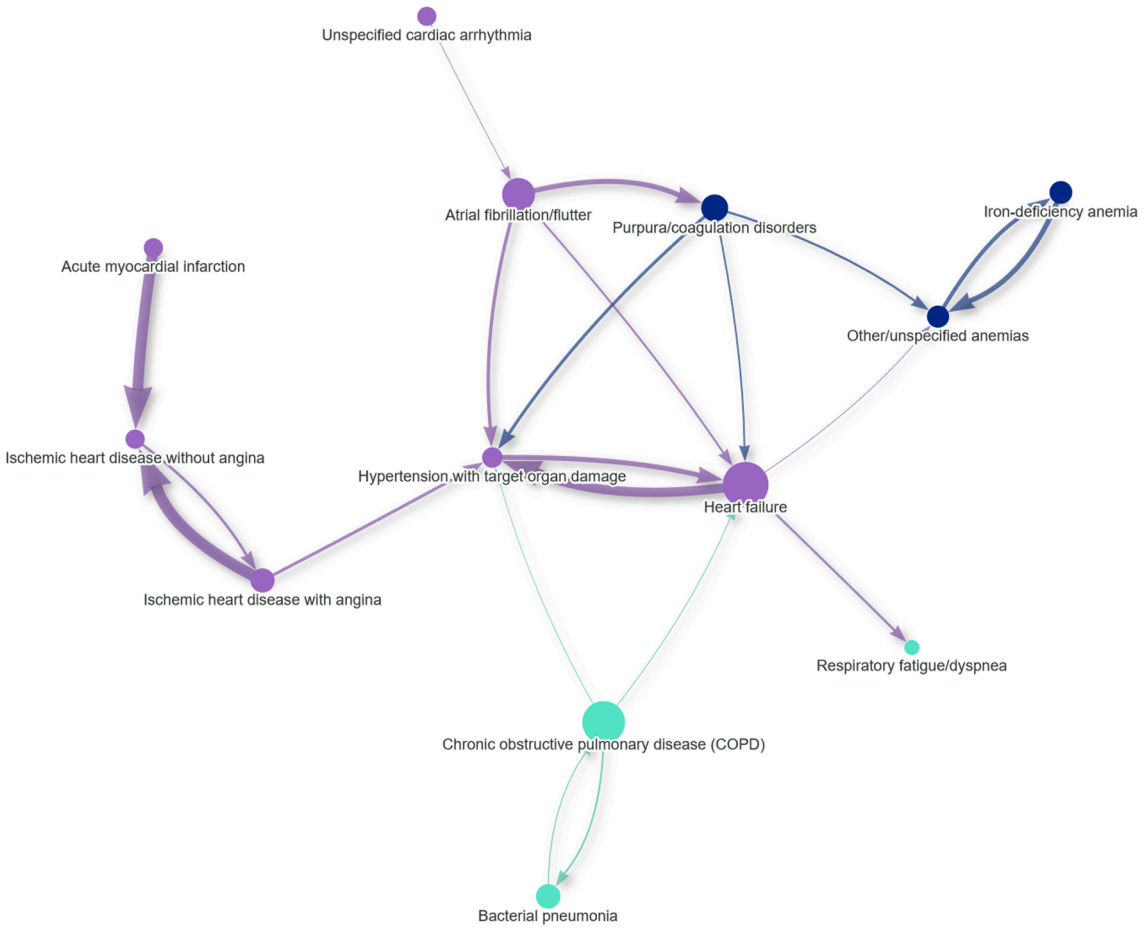


Fig. 1. Example of a graph generated by the platform. This is a subset shown in Fig. 6 below.

pair to be included in the study. This filtering reduced the number of code pairs to 2,588. After this filtering, we then proceeded to calculate the following incidence measures throughout the dataset (codification used for impact analysis):

- n_{diag1}: Number of patients diagnosed with only the first disease (without the second one)
- n_{diag2}: Number of patients diagnosed with only the second disease (without the first one)
- $n_{dir1\to2}$: Patients diagnosed with the first disease first, then the second disease
- $n_{dir2\to1}$: Patients diagnosed with the second disease first, then the first disease
- *Diagnosed together*: Patients diagnosed with both diseases at the same time (same appointment with the clinician)

With this information for each disease pair, analyses of the impact of one disease on the other one were performed.

3.3. Methods

3.3.1. Trajectory impact

To mathematically analyse which trajectories had the highest impact, different approximations such as the probability (p_{12}) or Cramer's V [19] were analysed. The Risk Increase (RI) has proven to be the most effective in achieving optimal graphical representation. This approach shifted the focus to ratios that are high relative to the incidence of the second disease. First, we define the probability:

$$p_{12} = \frac{n_{dir1\to2}}{n_{diag1}}$$

p_{12} represents the proportion of people who developed the second disease after having been diagnosed with the first one. Finally, to estimate RI_{12} we divided the probability (p_{12}) by the incidence of the second disease:

$$RI_{12} = p_{12} * \frac{n_{total}}{n_{diag2}}$$

This value represents the extent to which the likelihood of contracting the second disease is elevated after being diagnosed with the first disease.

3.3.2. Graphs

To effectively visualize the complex interactions between diseases and their likelihood of progression, we utilized the Pyvis Python software library [20], which enables the creation of interactive network graphs. In this way we obtain a graph similar to the one presented by Hjaltelin et al. [21], but providing more information and interactivity. See example graph in Fig. 1.

In our visualization, each node represents a disease. The size of the nodes corresponds to the number of individuals diagnosed with that disease.

The arrows connecting the nodes depict the significance found in the corresponding direction of disease trajectory. The width of the arrows represents the magnitude of the impact of the trajectory, according to the metric selected from those explained in the Error: Reference source not found section.

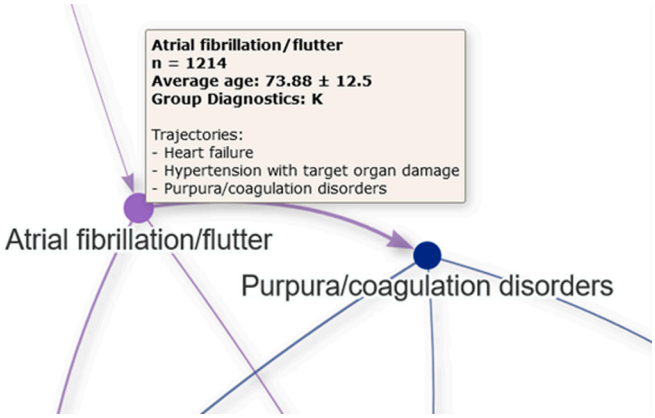


Fig. 2. Subset of Fig. 6 in html format, with the mouse over the atrial fibrillation node to show node hovering.

To enhance the interpretability of the graph, we incorporated a color-coding system derived from the BDCAP coding. Each disease node is assigned a specific colour based on the corresponding letter representing its group. This categorization aids in quickly identifying disease clusters or related conditions within the overall network.

The interactive nature of the visual representation allows clinical users to explore and interact with the graph. They can zoom in and out and scroll through the network. Additionally, hovering the mouse over the nodes or arrows displays information on disease characteristics and the dynamics of transitions between diseases within the cohort. The following breakdown provides the information displayed upon hovering as shown in the examples Fig. 2 and Fig. 3:

4. Results and clinical evaluation

This section outlines the key findings from our analysis. For the purposes of this presentation, we focus on four main cases. However, it is important to note that the framework is not limited to these cases, and the user can modify it to explore other specific scenarios.

Fig. 4 shows a graph including the 100 most significant trajectories. The disease names have been removed for the graph to improve readability, however, the original html format reveals several associations already documented in the literature, such as the development of dementia after a transient ischaemic attack [22,23,24]. The incidence of individuals developing dementia after a transient ischaemic attack is 4 %, which is 10.77 times higher than the normal incidence of dementia. It can also be seen that, on average, 2.21 (std 0.97) years elapse between the occurrence of the transient ischaemic attack and the diagnosis of dementia.

4.1. Graph by disease

Another option is to analyse a particular disease and arrange the graph based on its predecessor and successor diseases. This approach allows for a targeted analysis of a specific disease, which may be useful for newly diagnosed patients. Fig. 5 illustrates respiratory fatigue/dyspnoea trajectories, showcasing both preceding and subsequent diseases. The x-axis represents temporality when indicated.

4.2. Clinical analysis

Clinical evaluation by clinical experts has found that the most clinically relevant trajectories were those related to the dyspnoea. Fig. 5 shows the main pathologies that may lead to an increased diagnosis of dyspnoea. While dyspnoea is a symptom of heart failure, acute myocardial infarction, and asthma, its association with other conditions like osteoarthritis, diarrhoea, or coagulation disorders may not be immediately evident. Osteoarthritis and dyspnoea may coexist due to their higher prevalence in advanced age. Purpura may coexist with dyspnoea in severe situations, such as multiorgan failure. The link between diarrhoea and dyspnoea is not well understood, although it could be attributed to severe metabolic acidosis resulting from diarrhoea, leading to hyperventilation.

4.3. High prevalence graph

The graph in Fig. 6 shows the most prevalent trajectories; which require at least 30 people in the trajectory and a RI value greater than 5.

4.4. Clinical analysis

Exploring Fig. 6 from a clinical standpoint, we observe associations grouped by organs and systems. Some are interesting and logical, while others are self-evident such as heart-related diseases, coloured in purple.

The most interesting part of this graph lies in the cardiovascular and pulmonary zone, revealing the interconnection between the respiratory and the circulatory systems. It shows that patients with COPD have a higher risk of developing pneumonia, heart failure, and hypertension in this cohort. This aligns with medical knowledge confirming the cohort's

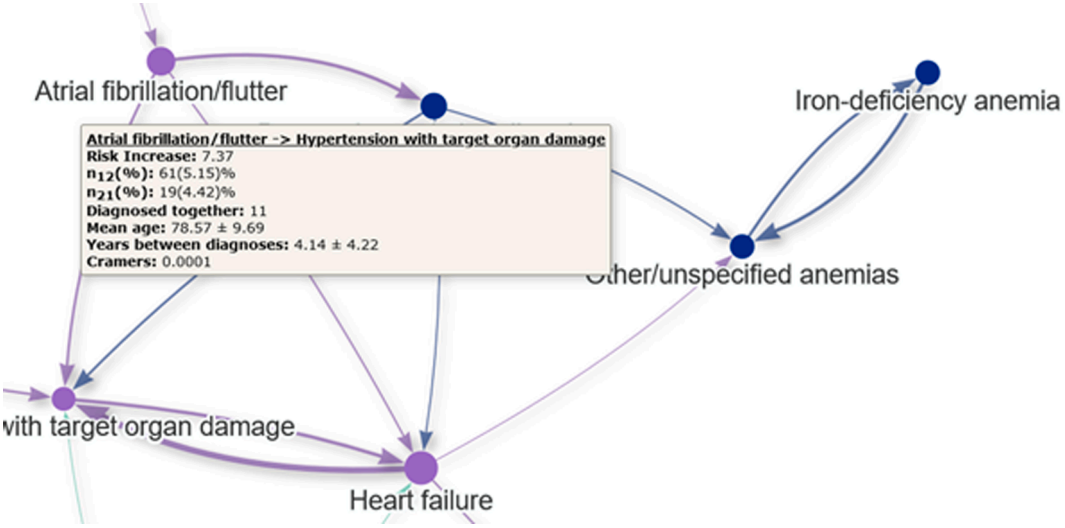


Fig. 3. Subset of Fig. 6 in html format, with the mouse over trajectory from atrial fibrillation hypertension with target organ damage to show the edge hovering.

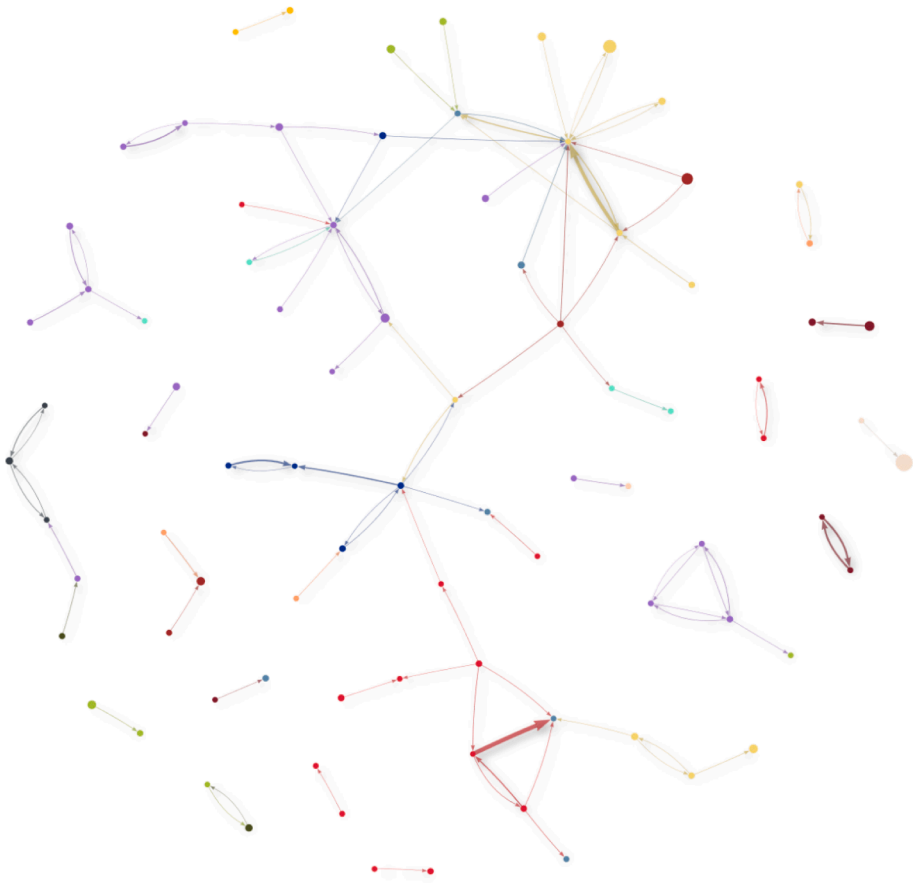


Fig. 4. The 100 most significant trajectories regarding the RI (represented by the thickness of the arrows). The name of each disease has been deleted as it is not legible at this size. The colour encode the typology of the diseases.

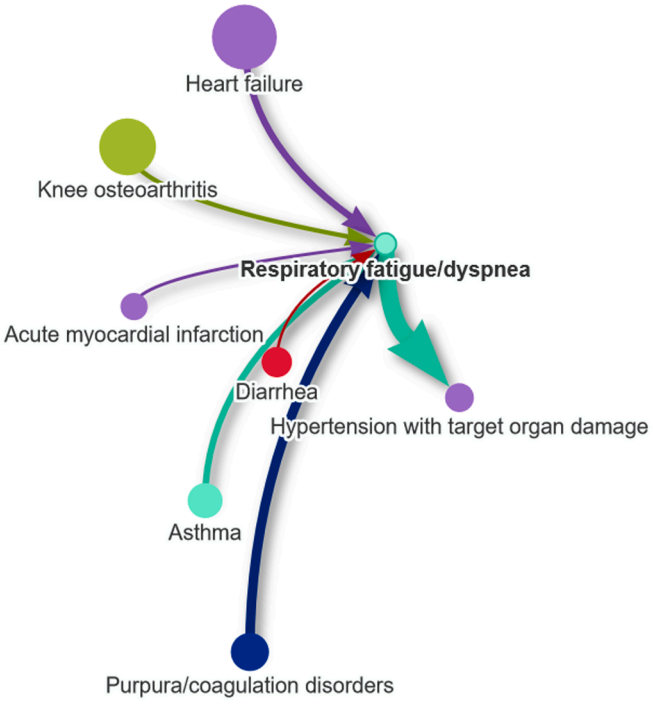


Fig. 5. Respiratory fatigue/dyspnoea-related trajectories. Those with an RI greater than 5.

adherence to established medical understanding.

It is intriguing to observe that the likelihood of facing complications from medical treatment increases after the diagnosis of rectal and colon neoplasms. This phenomenon might be attributed to the toxicity of the therapy used to treat these tumours, rather than the disease itself.

4.5. *High Risk Increase*

In Fig. 7, we present the trajectories with an RI above 15, indicating high significance.

4.6. *Clinical analysis*

From a clinical point of view, in Fig. 7 we observe, there are many cases where the arrows connect diagnoses that actually refer to the same condition: neuritis is a generic term for carpal tunnel syndrome, seizures are symptoms of epilepsy, non-specific arrhythmia is a non-specific diagnosis for paroxysmal tachycardia, urinary retention is a disorder of the urinary system, hiatal hernia is a disease of the oesophagus, and finally, non-specific anaemias can be haematologic neoplasms, as well as haematologic diseases. On the other hand, linking insulin-dependent diabetes with non-insulin-dependent diabetes does not make sense. It would make sense if it were in the opposite direction. This could be attributed to doctors' interchangeable use of similar diagnostic codes.

5. Discussion

The analysis of population trajectories in Tolosaldea (region of the Basque Country, Spain) provides valuable insights for policymakers in understanding the disease patterns within this region. The use of specific

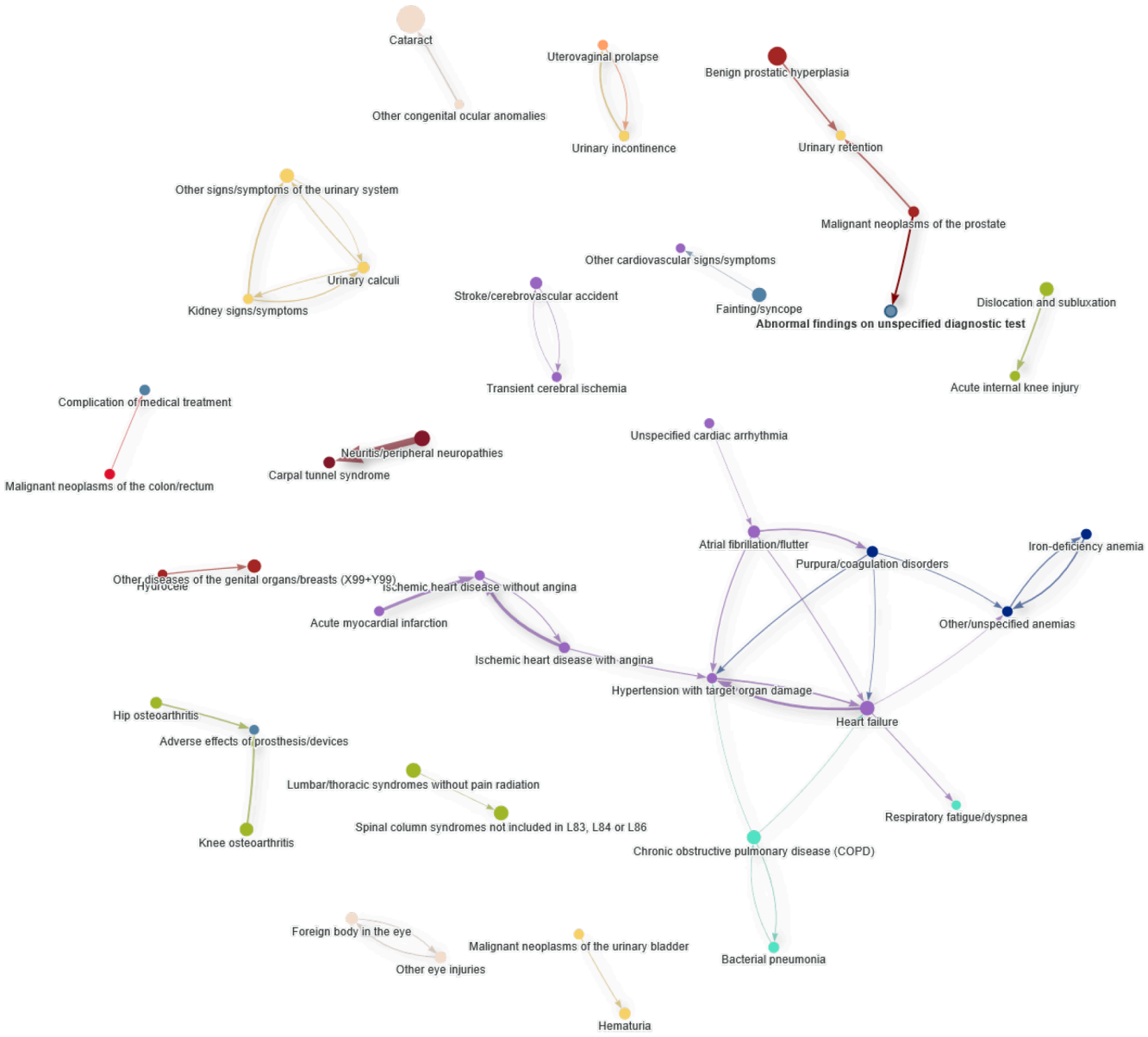


Fig. 6. This figure shows the disease trajectories of at least 30 patients and a minimum RI value of 5.

statistics and visual analysis has facilitated a comprehensive understanding of disease trajectories, enabling informed decision-making processes to be more relevant to the local population. Moreover, this analysis has empowered hospital clinicians to identify patterns specific to their patient population and healthcare facilities.

While several trajectories depicted in the graphs may already be known or intuitive, this is to be expected as the analysis is based on actual patient data. Using this methodology and studying this specific group, we have successfully identified the link between transient ischaemic attack and dementia which is supported by the state of the art [22,23,24] and many other clinically supported trajectories.

In this respect, an interesting avenue for further investigation would be to identify any notable trajectories that are absent from the analysis and to explore potential reasons for their exclusion. In other words, if there is a proven trajectory in the state of the art, and it does not occur in this population, it should be further investigated.

Despite the discovery of numerous supported trajectories, there exists a trajectory with high impact involving insulin-dependent diabetes and non-insulin-dependent diabetes that cannot be explained medically. It is imperative to investigate whether these data have been accurately collected, if there is a coding error, or if this case has been miscalculated.

A noteworthy limitation of this study is the relatively small sample size of the population under study. Ideally, population-based studies would benefit from larger sample sizes to enhance the generalisability of the findings. Additionally, since hospitals may have greater expertise in a particular specialty, they will have more of such patients. This adds noise to the trajectories, and it is possible that certain significant trajectories may not have been captured. However, this limitation does not impede the hospital-level analysis, as it can provide valuable insights within their specific context. For future research, it would be valuable to replicate this study with a larger and more diverse cohort, ideally at a national level, as demonstrated by similar studies conducted in Denmark [13 14]. This would enable the extraction of more general patterns that can benefit the entire population or to corroborate the relationships observed in this study.

The sample size also constrains the feasibility of conducting a two-step analysis of diseases. In simpler terms, if two diseases are linked through another disease, we are unable to draw conclusions since the study does not encompass an examination of the combined impact of these two diseases. A more extensive cohort would provide the

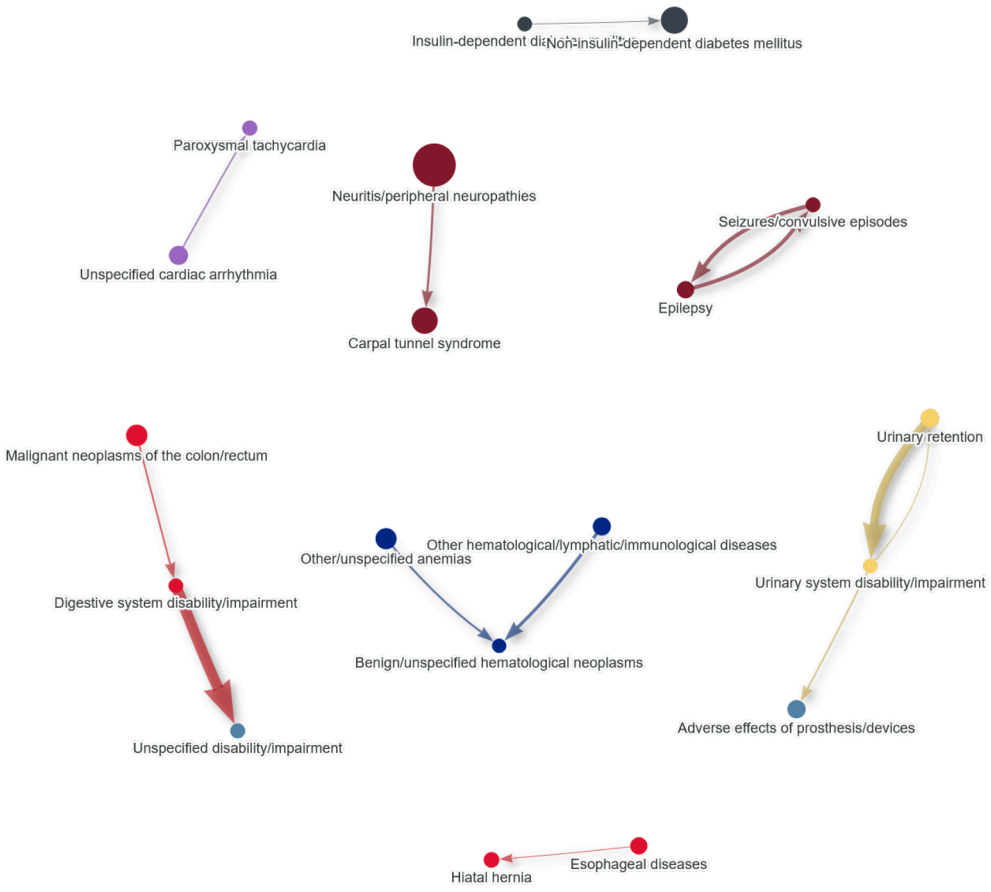


Fig. 7. This graph shows the trajectories with a high impact, i.e., RI higher than 15.

opportunity to explore the impact of the combination of two diseases.

Similarly, future work should focus on implementing AI models to predict the likelihood of developing a second disease more accurately.

Despite the acknowledged limitations, this study has successfully generated a summary of disease trajectories that can be easily interpreted. The identification of valuable information and patterns from these trajectories supports the discovery of meaningful insights.

6. Conclusions

In conclusion, this article presents the findings of a trajectory analysis conducted on a cohort of 71,849 patients, utilizing trajectory impact analysis and interactive visual methods. Several methods have been proposed to estimate the impact of disease trajectories, with the Risk Increase (RI) being the one that best describes the trajectories according to clinicians. The analysis focused on providing an approach for examining disease trajectories and identifying key points in disease progression.

Some relevant trajectories were found. Some are obvious, such as the trajectories between epilepsy and convulsions, and others are less obvious, but supported by the state of the art, such as dementia after transient ischaemic attack and stroke.

In summary, a simple and easy-to-understand method of analysing disease trajectory data has been presented. This provides clinicians with valuable information to guide management decisions.

7. Summary table

What was already known on the topic:

1. Several studies have been conducted analyzing millions of patients and their trajectories (6 million and 10 million patients).
2. An interface has been created for the exploratory analysis of disease trajectories.
3. Trajectories have been analyzed using various statistical methods.

What this study added to the knowledge:

1. A metric (Risk Increase) has been proposed to measure the impact of disease trajectories.
2. Interactive graphics have been proposed for a detailed description of different disease trajectories.
3. Statistical methods and visual analytics methods have been combined for a comprehensive analysis of disease trajectories within an entire cohort.
4. A database of 71,849 patients has been exploited for the extraction of disease trajectories, previously unused for this purpose. Extracted trajectories match with clinical know-how, validating the approach.
5. A way to harmonize and group ICD 9 and 10 codes has been proposed, enabling the comparison between them.

Funding

This study was supported by the grant ZL-2023/00403 Hazitek, called DISTRA, from Eusko Jaurlaritza (Basque Government) and the European Union under the European Regional Development Fund (ERDF).

Credit authorship contribution statement

Jon Kerexeta-Sarriegi: Writing – review & editing, Writing –

original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Teresa García-Navarro:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **María Rollan-Martinez-Herrera:** Validation. **Nekane Larburu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Moisés D. Espejo-Mambié:** Validation, Methodology. **Andoni Beristain Iraola:** Writing – review & editing, Supervision. **Manuel Graña:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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